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Your Knowledge... Our Solution...!



FT-NIR SPECTROSCOPY

TANGO

BRUKER

Faster, simpler, more secure - with TANGO your NIR analysis speeds up.

Innovation with Integrity

Forget tedious training courses or specialized knowledge. TANGO makes it simple for you: with its intuitive user interface it guides the operators quickly and securely through the work flow. Also untrained staff can operate the spectrometer securely and error free. Good to know: the user interface is available in 17 languages.

Logical: Sample handling the easy way

When it comes to sample handling, TANGO saves time and prevents operating errors. No sample preparation is necessary, no chemicals, no additional reagents. The sample is simply filled into the measuring cup or vial with the measurement being done through glass.

Automatic: Background measurements

Contributing to the high precision of FT-NIR spectrometers is the exact alignment with the background. TANGO goes one step further: The background measurements are performed automatically without user intervention. This alignment can even take place if a sample is located in the sampling position. The perfect prerequisite for optimal and secure measurements at any time – without (human) errors.

TANGO has exactly what users require of an FT-NIR spectrometer suitable for industrial use: Robustness, high precision and straight-forward operator guidance.

- Proven FT-NIR technology by Bruker
- Easy-to-use touch screen operation
- Automated background management
- Small footprint
- Robust and precise optics
- 10 years of warranty on the moving parts of the interferometer
- Direct calibration transfer to and from all existing Bruker Optics NIR spectrometers
- Data exchange via network
- Complies with 21 CFR Part 11 requirements

TANGO for all applications

The TANGO offers an efficient and cost-effective analysis for material identification and quantification of constituents in a wide variety of application fields.
The most important areas of application cover:

Food and everage Industry

Feed Manufacturing

Pharma and Biotechnology

Chemical and Petrochemical

Polymer Industry

Ergonomic: User comfort in all ways.

Nowadays, with laboratory space being at a premium, it is difficult to locate a the new analyzer in the lab. This isn't an issue with TANGO due to its small footprint and ability to be positioned in any orientation - it can be operated from all three sides. The monitor swivels in all directions and can even be tilted in various positions to enable an optimum view. Alternatively, it can be placed flat on top of the spectrometer for a perfect viewing from above.

Practical: Efficiency in every detail.

TANGO demonstrates how easy NIR analysis can be, not just during measurements. The housing and touch-screen monitor are rugged and easy to clean – ideal for lab and at-line environments. The exchange of the light source, if required, can be done in a few seconds. Also the desiccant cartridge, which keeps the inside low on humidity can be changed easily.

Transmission measurements: highest precision for liquids.

TANGO for measuring liquids comes equipped with an active sample heater/cooler as standard. The temperature of the sample can be selected between +20°C and +80°C. A sensor is permanently checking the temperature of the sample vial. This allows for a fast temperature control of the sample and therefore quick and reliable results.

Reflection measurements: analyze solids efficiently.

TANGO for measuring solids contains a gold coated integrating sphere for analysis by diffuse reflection. This sphere ensures highly reproducible measurements of inhomogeneous samples. The diameter of the measurement spot is around 10 mm; the measured sample surface can be enlarged by using sample rotators.



The results can be easily read from the clearly structured graphical interface.



TANGO is also available without integrated PC and touch-screen monitor for a seamless integration into existing IT periphery.



Personal consultation and customer service guarantee a sustainable and efficient solution.

Bruker Optics is continually improving its products and reserves the right to change specifications without notice.
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Standards and Approvals - CE labeled and complies with the following directives:
EMC Directive 2014/30/EU, Low Voltage Directive 2014/35/EU, WEE Directive 2012/19/EU, RoHS II Directive 2011/65/EU, RoHS III 2015/863/EU
Laser class 1 product.

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Bruker Optics is ISO 9001, ISO 13485, ISO 14001 and ISO 50001 certified.

Process Spectroscopy
www.spectroscopy-in-process.com

Online information
www.bruker.com/TANGO



TANGO FT-NIR Spectrometer for Reflection Measurements

Specifications

Rugged NIR analyzer with sealed and desiccated optics for routine quality control. The analyzer is equipped with a gold coated integrating sphere for the reflection measurement of samples. Sample spinners for inhomogeneous samples are optionally available.
The analyzer is either equipped with a built-in data system with a touch screen or optionally available with an external PC-based data system. The easy-to-use and GMP compliant TANGO software package allows qualitative and quantitative analysis.

Design

■ Housing:	rugged, sealed and desiccated housing
■ NIR source:	air cooled Tungsten source, average lifetime 9,000 h, pre-aligned, easily exchangeable by operator
■ Interferometer:	RockSolid™, frictionless bearing, permanently aligned, Quartz substrate beamsplitter with proprietary coating 10 years warranty on the moving parts of the interferometer.
■ Laser:	Solid State Laser (10 years warranty)
■ Detector:	InGaAs diode, TE-cooled and temperature-stabilized
■ Sample presentation:	reflection measurements in cups, Petri dishes or disposable vials background measurement w/o removing the sample

Performance

- Spectral range: 11,500 - 4,000 cm^{-1}
- Resolution: better than 4 cm^{-1}
- Wavenumber reproducibility: better than 0.04 cm^{-1}, at 7,181 cm^{-1}
- Wavenumber accuracy: better than 0.1 cm^{-1}, at 7,181 cm^{-1}
- Photometric accuracy: better than 0.1% T

Electronics

- Data acquisition: 24-bit A/D converter
- Automation: microprocessor controlled optical bench, digital speed control, automatic gain selection, internal background, advanced system check
- Performance check: permanent on-line diagnostics of all optical components, automation units and sampling accessories
- Connectivity: Ethernet interface, 10/100 Mbps

Dimensions

- Spectrometer (w x d x h): 309 mm x 438 mm x 175 mm (plus max. 250 mm height for monitor)
- Weight: 22.0 kg (17.5 kg w/o data system)

Operating Environment

- Degree of protection: IP53 (TANGO with external PC)
- Operating temperature: 5° C to 35° C
- Storage temperature: -20° C to 70° C
- Operating relative humidity: < 80% non-condensing
- Power requirements: 100 to 240 VAC, 50/60 Hz, 75 W (typical 40 W)
- Regulatory certification and compliance: CE, WEEE and RoHS

Data System

- Embedded PC (1,86GHz, 4GB RAM, 64GB SSD) with 12.2" touch display (capacitive); optional external PC data system

Validation

- TANGO software: the software package includes functions that facilitate the use of the analyzer in a 21 CFR Part 11 compliant environment
- Instrument qualification: TANGO software supports Operational Qualification (OQ) and Performance Qualification (PQ) optional: qualification according to USP1119/PhEur 2.2.40
- Internal validation unit: filter wheel with reference standards, for automatic PQ tests
- Validation manual: complete hardware and software validation documentation is available as an option
- Service contracts: Preventive maintenance contracts and validation services are available as an option

Software

- GMP compliant TANGO software with user-friendly interface
- Product based operation (Barcode reader friendly)
- Qualitative and/or quantitative evaluations (no method development)
- Customizable analysis reports
- Data exchange via USB stick and/or LAN



Product selection by pressing the respective icon



After the measurement the results are displayed graphically

Standards and Approvals

CE labeled and complies with the following directives:

- EMC Directive 2014/30/EU
- Low Voltage Directive 2014/35/EU
- WEE Directive 2012/19/EU
- RoHS II Directive 2011/65/EU
- RoHS III 2015/863/EU

Bruker Optics is ISO 9001 and ISO 13485 certified.

Laser class 1 product.

Covered by one or more of the following patents: DE102004025448; DE19940981.
Additional patents pending.

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Application Note AN N287

FT-NIR Analysis of Feed

The cost of animal nutrition represents the largest operating cost for most commercial livestock producers. In order to maintain an optimum balance between feed costs and productivity, all feed ingredients should be analyzed for nutrient concentration and these values are then used to formulate the rations and required supplements. However, the included feedstuffs vary widely in composition, due to origin, seasonal changes or year. Fourier Transform Near Infrared spectroscopy (FT-NIR) provides a fast and effective solution for analyzing raw materials as well as finished feeds in order to optimize the production steps and monitor the final product quality.

Easy Sample Analysis with FT-NIR

Bruker Optics offers the most comprehensive range of FT-NIR solutions for quality control and formulation adjustments. Samples can be analyzed non-destructively in seconds, saving costs by reducing time and reagent use. Analyzing by FT-NIR in the lab or at-line close to the production requires just filling an easy-to clean cup with the solid sample and presenting it to the analyzer. Liquid samples like oils or molasses can be analyzed in disposable vials with the same spectrometer.

Ready to use Calibration Packages

A set of universal FT-NIR calibrations for the analysis of finished products in the feed industry are available. These calibrations help you to achieve a superior quality control, leading to an enhanced performance of your products.

FT-NIR spectroscopy offers a rapid, accurate and non-destructive tool for the analysis of different types of:

- Ruminant Feed (Calf, Dairy & Beef Cattle)
- Poultry Feed (Chick, Broiler, Breeder, Layer)
- Pig Feed (Piglet, Weaner, Grower, Finisher, Sow)
- Horse Feed
- Aqua Feed

on various parameters like

- Moisture
- Fat
- Protein
- Fiber
- Ash

The following more specialized parameters are available if applicable:

- Starch
- Minerals (Ca, Na, K, P)

Performance Statistics Poultry Feed

Property		Data Set			Calibration		Validation	
Name	Unit	n	Min	Max	R^2	RMSEE	R^2	RMSEP
Ash	%	1871	1.0	14.5	77.6	1.07	68.3	1.36
Fat	%	3280	0.5	24.1	95.2	0.50	95.7	0.49
Fibre	%	2402	0.1	27.7	94.7	0.90	94.5	0.70
Moisture	%	4127	4.6	16.3	88.2	0.48	81.2	0.42
Protein	%	5058	4.0	47.4	95.3	0.97	92.0	0.84
Starch	%	1217	4.8	57.3	98.9	1.09	95.2	1.37

Performance Statistics Ruminant Feed

Property		Data Set			Calibration		Validation	
Name	Unit	n	Min	Max	R^2	RMSEE	R^2	RMSEP
Ash	%	459	2.9	15.0	85.7	0.86	83.5	0.91
Fat	%	1454	1.5	18.0	95.0	0.47	89.6	0.37
Fibre	%	347	0.8	16.8	96.8	0.76	92.6	1.07
Moisture	%	1999	5.3	17.0	88.3	0.40	86.4	0.43
Protein	%	2628	8.1	39.1	92.8	0.87	92.2	0.78
Starch	%	1467	0.6	58.6	97.9	1.50	97.8	1.60

FT-NIR Spectrometers: Bruker Optics offers various FT-NIR spectrometer models for lab, at-line and on-line applications:

TANGO



FT-NIR analyzer for routine use in the lab.

MPA II



Multi Purpose Analyzer for maximum flexibility.

MATRIX-I



At-line analysis with optional NEMA4/IP66 protection.

MATRIX-F



Process monitoring with probes and measurement heads.

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Application Note AN N285

FT-NIR Analysis of Feed Ingredients

The cost of animal nutrition represents the largest operating cost for most commercial livestock producers. In order to maintain an optimum balance between feed costs and productivity, all feed ingredients should be analyzed for nutrient concentration and these values are then used to formulate the rations and required supplements. However, the included feedstuffs vary widely in composition, due to origin, seasonal changes or year. Fourier Transform Near Infrared spectroscopy (FT-NIR) provides a fast and effective solution for analyzing raw materials as well as finished feeds in order to optimize the production steps and monitor the final product quality.

Easy Sample Analysis with FT-NIR

Bruker Optics offers the most comprehensive range of FT-NIR solutions for quality control and formulation adjustments. Samples can be analyzed non-destructively in seconds, saving costs by reducing time and reagent use. Analyzing by FT-NIR in the lab or at-line close to the production requires just filling an easy-to clean cup with the solid sample and presenting it to the analyzer. Liquid samples like oils or molasses can be analyzed in disposable vials with the same spectrometer.

Ready to use Calibration Packages

A set of universal FT-NIR calibrations for the analysis of a wide range of raw materials in the feed industry are available. These calibrations help you to achieve a superior quality control, leading to an enhanced performance of your products.

FT-NIR spectroscopy offers a rapid, accurate and non-destructive tool for the analysis of:

- Moisture
- Fat
- Protein
- Fiber
- Ash

The following more specialized parameters are available if applicable:

- Starch
- Total Sugar
- ADF
- NDF

In addition Bruker FT-NIR instruments are supported by 3rd party suppliers of more specialized calibration packages and of special services, e.g. analysis of Amino Acids in feed ingredients.

List of Feed Ingredients

Cereals

Barley
Corn/Maize
Rice
Sorghum
Triticale
Wheat

Cereal By-products

Corn/Maize
- Gluten Feed
- Gluten Meal

Wheatfeed (Midds)

Oil Seeds and By-products

Rapeseed/Canola
- Expeller
- Meal
- Unground/whole

Soybean
- Expeller
- Ground
- Meal

Sunflower
- Expeller
- Ground
- Meal

Animal Proteins

Blood Meal
Bone Meal
Feather Meal
Fish Meal
Meat and Bone Meal
Poultry By-product Meal

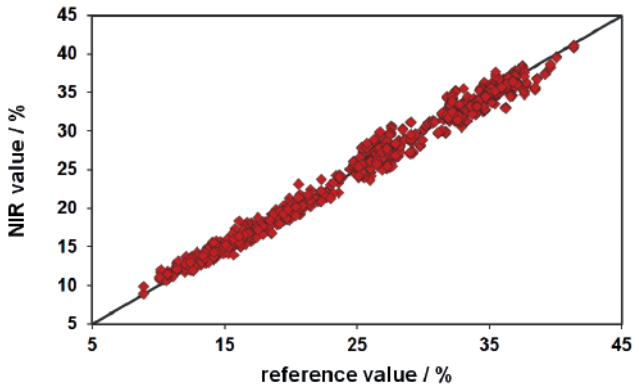
Other By-products

Biscuit Meal
Distillers Grains (DDGS)

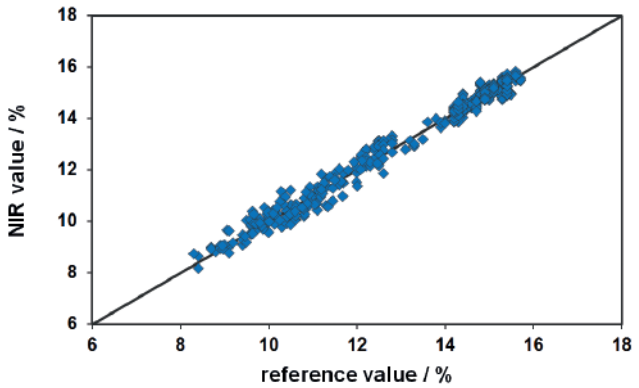
Bruker Optics strives to continuously optimize its portfolio of calibration methods. Therefore additional products may already be available on request.

Validation Results for Various Ingredients

Sunflower: Protein Content



Wheat: Moisture Content



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General Parameters analysed using Bruker FT NIR:-

Solid/Powder/Gel/Seeds/Viscous Samples:-

- Crude Protein/Protein
- Fat/Oil/Ether Extract
- Moisture
- Crude Fiber
- Starch
- Lactose
- Broken Starch
- Semolina
- Sugar (Total and individual)
- Color
- Metabolizable Energy/Digestibility energy
- ADF and NDF
- Fatty acids
- Free Fatty acids
- Erucic acid
- Glucosinlates
- Viscosity
- Gluten-Dry/Wet
- **Amino acids :-**
    - Methionine
    - Cystine
    - Met + Cys.
    - Lysine
    - Threonine
    - Tryptophan
    - Arginine
    - Isoleucine
    - Leucine
    - Valine
    - Histidine
- +Phenylalanine
- Salt
- Iodine
- TVBN
- Falling Number
- Phosporous
- Sand & Silica -Total Ash
- pH
- Curcumin.
- Piperine.
- Capsalicin

- **In Molasses:-**
  - Sucrose,
  - POL,
  - BRIX,
  - Purity.
  - Invert Sugar
  - Total Sugar
- Soya,Sunflower meal-UDP….etc.

Parameters for Oil Samples:

- Anisidine Value
- TBHQ
- Dropping Point
- Moisture
- Gardner Color
- Lovibond LY
- Lovibond R
- Iodine Value.
- TFA Trans Fatty Acids
- FFA Acidity
- Anisidine Value
- Peroxide Value
- Solid Fat Determination
- Saponification
- Unsaponification
- pAV
- EPA (Eicosapentaenoic Acid)
- DHA (Docosahexaenoic Acid)
- DPA (Docosapentaenoic acid)
- SDA (Stearidonic acid)
- Total Omega-3 Content
- **Free Fatty Acids-Profile:-**
  - C16:0 (Palmitic Acid)
  - C18:0 (Stearic Acid)
  - C18:0 (Stearic Acid)
  - C18:1 (Oleic Acid)
  - C18:2 (Linoleic Acid)
  - C18:3 (Linolenic Acid) etc…



TANGO FT-NIR spectrometer







Rapid Feed and Petfood Analysis with FT-NIR Spectroscopy

Bruker offers the award-winning TANGO FT-NIR spectrometer exclusively for the Indian market dedicated to the Feed,Food, Oil, Pharma industry:

- **Maintain optimum balance between feed costs and productivity by analyzing feed and feed ingredients.**
- **Monitor feed products by checking quality of incoming feed ingredients and finished feeds.**
- **Invest into the future! Bruker FT-NIR allows direct calibration transfer across all models and even to online measurement systems.**
- **User friendly, easy to use, fast, simple, more secure - with TANGO for accurate and reilable proximate analysis of all variety of samples. Forget tedious training courses or specialized knowledge.**
- **TANGO makes it simple for you: with its intuitive user interface it guides the operators quickly and securely through the work flow.**

FT-NIR Innovation with Integrity

- Proven FTNIR RockSolid Interferometer technology with permanently aligned Gold coated optics
- Automated background management
- Laser and interferometer with 10 years lifetime warranty
- Robust and precise optics
- Direct calibration transfer to and from all existing Bruker Optics NIR spectrometers and other vendors
- Data exchange via network

Easy Sample Analysis

Samples can be analyzed non-destructively in seconds, saving costs by reducing time and reagent use.

Liquid samples like oils or molasses can be analyzed in glass vials with the same spectrometer.

Also untrained staff can operate the spectrometer securely and error free. The sample is simply filled into the measuring cup or vial with the measurement being done through Quartz glass bottom. The housing is rugged and easy to clean – ideal for lab and at-line environments.

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